Fiscal Year:	FY 2011	**Task Last Updated:**	FY 07/12/2011
PI Name:	Collins, Cynthia Ph.D.		
Project Title:	Gravitational effects on biofilm formation during spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	12180-3522	**Congressional District:**	21
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	12/15/2009	**End Date:**	05/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-132 ; STS-135 ; ISS NOTE: Extended to 5/31/2014 per A. Chu/ARC and NSSC information (Ed., 5/22/13) NOTE: Extended to 5/31/2013 per A. Chu/ARC and NSSC information (Ed., 6/1/2012) NOTE: to be extended per A. Chu/ARC (Ed., 1/12/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute) Dordick, Jonathan (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX09AI70G-Flight portion		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	There is an urgent need to understand the effects of microgravity on the growth, cellular physiology, and cell-cell interactions in microbial biofilms. This information can then be used to curtail harmful activities of microbial consortia thriving as biofilms on the International Space Station and for the long-term success of human space exploration. Bacterial biofilms were abundant on the MIR space station and were responsible for increasing corrosion and blocking a water purification system. Health and safety hazards linked to the development of biofilms are also of great concern, including increased rates of infection due to the enhanced survival of organisms within biofilms and increased resilience of biofilm populations to antimicrobial compounds. Furthermore, experiments with Escherichia coli have indicated that biofilms formed under microgravity may have even higher levels of resistance to antimicrobials. We propose to study how gravity alters biofilm formation with the goal of developing new strategies to reduce their impact on the operation of spacecrafts and the health of their crew. We will focus on two model microorganisms that form biofilms both inside and outside of the human body, Pseudomonas aeruginosa and Staphylococcus aureus. These microbes can switch between benign and pathogenic interactions with humans and may be relevant to crew health during extended missions. This project consists of a set of experiments to study biofilm formation during spaceflight that were conducted on Space Shuttle flight STS-132, which launched in May 2010. We examined how the formation and the three-dimensional structures of biofilms formed by these microbes differ in spaceflight compared to normal gravity. We will employ newly developed, non-leaching, nanotechnology-based, antimicrobial surfaces to aid in assessing differences between biofilm formation and structure in normal and microgravity and to assess their potential to reduce the impact of biolfims in future spacecraft designs. NOTE that this is the Flight portion of project entitled, "Gravitational effects on quorum sensing and biofilm formation."
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	According to the Center for Disease Control (CDC), hospital-acquired infections are the fourth leading cause of death in the United States behind stroke, cancer and heart disease. Furthermore, it is estimated that >65% of all bacterial infections are associated with biofilms. A greater understanding of biofilms is essential if we are to find effective methods to combat their formation. Furthermore, the low-shear conditions microbes experience in microgravity are similar to those found in the human body that are difficult to study. This work may provide new insights into the role of shear and other physical effects, such as convection, on biofilm formation.
Task Progress:	Our core hypothesis is that the multicellular behavior of bacteria differs under normal and microgravity conditions. Understanding the differential responses and physical effects of reduced gravitational force on biofilm formation may provide new insights into combating biofilm formation in space. Our secondary hypothesis is that our novel, nanotechnology-based biocatalytic surfaces can be optimized to inhibit biofilm formation during spaceflight. Prior to our recent work, several observational papers had described the formation and detrimental effects of biofilms to spacecrafts and space stations; however, only a single publication described the formation of biofilms in simulated microgravity. Lynch et al., developed a system for examining biofilm growth in simulated microgravity using glass microcarrier beads as substrates for biofilm formation 24. They successfully showed that E. coli forms biofilms in low-shear microgravity. In May 2010, we had the opportunity to fly a set of experiments in STS-132 designed to assess biofilm formation by Pseudomomas aeruginosa and Staphylococcus aureus during spaceflight. The results from our STS-132 flight experiment, known as Micro-2, have demonstrated that biofilm formation by both of these model organisms can occur during spaceflight. More importantly, we have demonstrated that these organisms respond differently to the spaceflight environment. Our spaceflight experiments (Micro-2) used specialized hardware designed by Bioserve Space Technologies (University of Colorado) known as a fluid processing apparatus (FPA). During Micro-2, we also employed BioServe's Commercial Generic Bioprocessing Apparatus (CGBA) and were able to modulate the temperature throughout the experiment. Our Micro-2 experiments compared biofilm growth by P. aeruginosa PA14 and S. aureus SA113 between normal gravity and spaceflight. Inoculum, media, and fixative (microscopy and gene expression samples only) were loaded into chambers of FPAs, separated by rubber septa. The media used for was an artificial urine medium (AUM) with 5 mM or 50 mM phosphate for P. aeruginosa, or tryptic soy broth (TSB) with 2.5% w/v sodium chloride for S. aureus. The substrate for biofilm formation, a mixed cellulose ester membrane, was included in the chamber with the media. The membranes used to test the role of anti-fouling molecules in preventing biofilm formation in microgravity were coated with a latex paint and anti-fouling enzymes covalently linked to multiwalled carbon nanotubes (MWNTs). The enzymes tested for their efficacy against P. aeruginosa were proteinase K and alginate lyase. An enzyme capable of breaking open and killing S. aureus cells, known as lysostaphin, was tested for its efficacy against S. aureus biofilms. Membranes coated with only paint containing MWNTs were used as a control. To initiate cell growth, inocula were combined with the media. Samples were incubated at 37°C for 72h then stored at 8°C until they were processed. Ground controls were conducted in parallel at Kennedy Space Center. For direct measurement of biomass, colony-forming units were determined using standard methods. Image acquisition was performed using a confocal laser-scanning microscope (CLSM) of paraformaldehyde fixed cells stained with propidium iodide. Images were analyzed using COMSTAT. RNA isolation and gene expression studies from Micro-2 are ongoing. Key results – Pseudomonas aeruginosa: 1. Spaceflight was observed to increase planktonic P. aeruginosa biomass by approximately 10-fold, but only in AUM with 5 mM phosphate. 2. Direct measurements of biomass showed P. aeruginosa PA14 formed 3 to 5-fold more biofilm during spaceflight compared to normal gravity. In addition, quantitative analysis of CLSM images showed increased biofilm biomass ($p<0.001$) and increased biofilm thickness ($p<0.001$) in samples grown during spaceflight. A clear effect of phosphate on biofilm biomass was not observed in any biofilm samples. 3. Biofilm formation by both the flgK and motABCD mutants was similar to wild type in normal gravity. However, unlike wild type spaceflight did not cause an increase in biofilm biomass by the motility mutants. 4. A significant decrease in biofilm formation was observed both with the coated membranes containing MWNTs and with the addition of alginate lyase. Additional controls are needed to separate the role of the paint, MWNTs and enzyme in the reduction

	of biofilm biomass. These controls and additional methods are being used to test and optimize the use of such coatings in microgravity. Key results – Staphylococcus aureus: 1. Spaceflight was not observed to affect S. aureus planktonic biomass. 2. Quantitative analysis of CLSM images showed decreased biofilm biomass ($p<0.05$) in samples grown during spaceflight. This data is from only 2 biological replicates due to contamination of a S. aureus inoculum. Recent data from ground-based experiments using a HARV have shown similar results (data not shown). 3. The membrane coatings did not appear to inhibit biofilm biomass formation. In normal gravity only, both the paint/MWNT and paint/MWNT/lysostaphin combinations were observed to decrease planktonic biomass. Unlike alginate lyase, lysostaphin is a microbicide and was expected to kill cells. The decrease in biomass with only latex paint and MWNTs was unexpected. Overall, we have clearly shown that biofilm formation by both P. aeruginosa and S. aureus is affected by microgravity. The observation that one showed an increase in biomass, while the other showed a decrease has allowed us to generate new hypotheses that are being tested in our ground-based experiments and in a second set of space-flight experiments.
Bibliography Type:	Description: (Last Updated: 04/16/2020)
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." 2010 AIChE Annual Meeting, Salt Lake City, Utah, November 7-12, 2010. 2010 AIChE Annual Meeting Conference Proceedings, 2010. http://www3.aiche.org/proceedings/Abstract.aspx?ConfID=Annual-2010&GroupID=1515&SessionID=13969&PaperID=203654 , Nov-2010
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." Poster presentation at the Biofilms-IV International Conference, Winchester, England, September 1-3, 2010. Biofilms-IV International Conference Proceedings, 2010. p. 89. http://biofilms4.files.wordpress.com/2009/06/biofilms-handbook-final-1-pq.pdf , Sep-2010
Abstracts for Journals and Proceedings	Kim WS, Tengra FK, Young Z, Shong J, Pangule R, Plawsky JL, Dordick JS, Collins CH. "Gravitational effects on biofilm formation by Pseudomonas aeruginosa." Oral Presentation at the 26th Annual Meeting of the American Society for Gravitational and Space Biology (ASGSB), National Harbor, MD, November 4-7, 2010. Program and abstracts. 26th Annual Meeting of the American Society for Gravitational and Space Biology (ASGSB), November 2010. p. 25. , Nov-2010